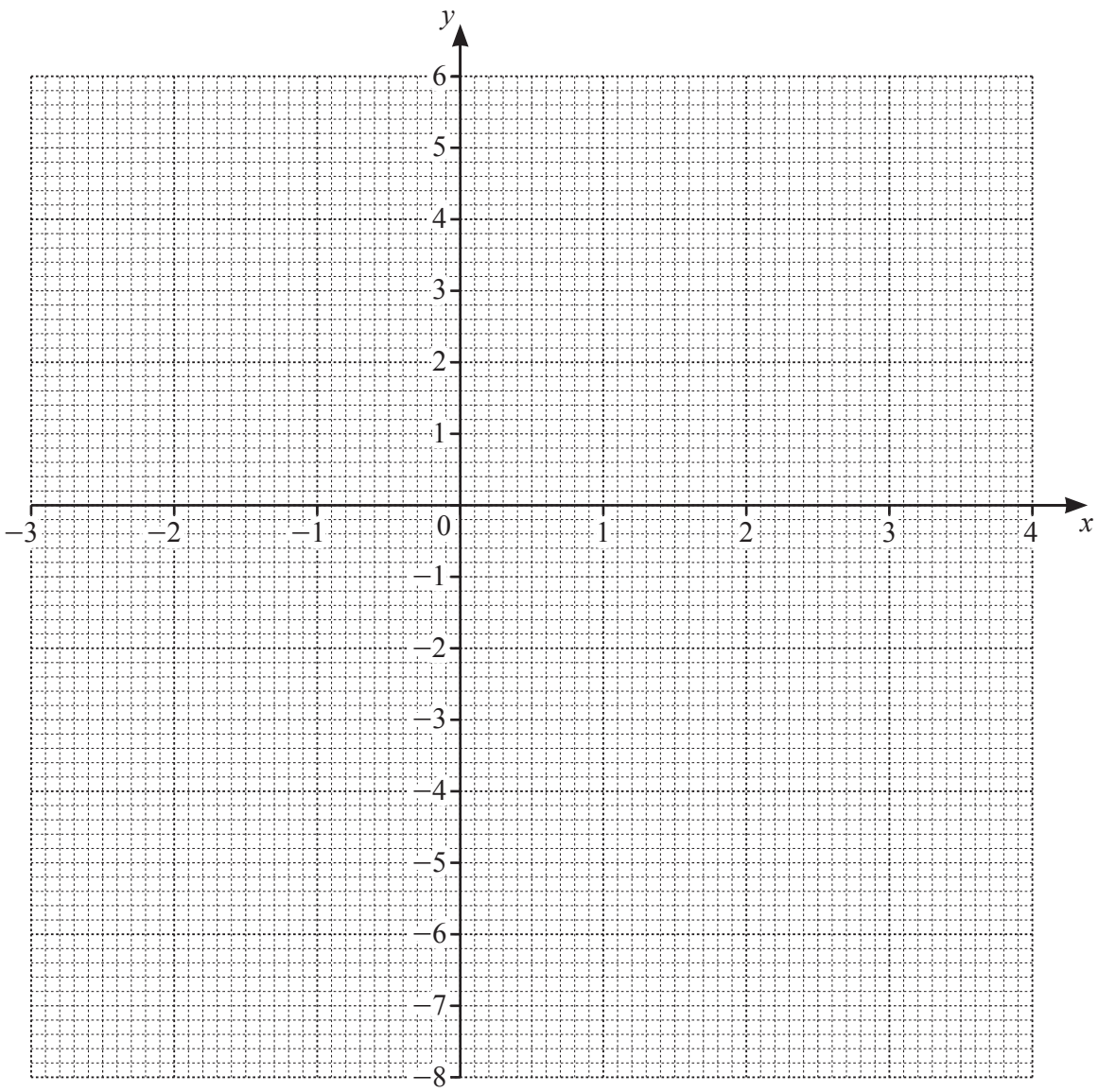
0580_s20_qp_33_Q8

8 **(a)** Complete the table of values for $y = -x^2 + x + 5$.

x	-3	-2	-1	0	1	2	3	4
y		-1	3			3		

[3]

(b) On the grid, draw the graph of $y = -x^2 + x + 5$ for $-3 \leqslant x \leqslant 4$.



[4]

0580_s20_qp_33_Q8

(c) Write down the coordinates of the highest point of the graph.

(..... , ...) [1]

(d) Write down the equation of the line of symmetry of the graph.

..... [1]

(e) **(i)** On the grid, draw the line $y = x$ for $-3 \leqslant x \leqslant 4$. [1]

(ii) Write down the values of x where the line $y = x$ crosses the curve $y = -x^2 + x + 5$.

$x =$ ..... and $x =$ ..... [2]